



УНИВЕРСИТЕТ
ИННОПОЛИС

FaceGuard V2

Team 16

Problem

When you approach the door with your hands full, it is inconvenient to take out the card in order to unlock it and open it yourself. Also, when someone approaches the door without a pass, human intervention is needed to open the door.

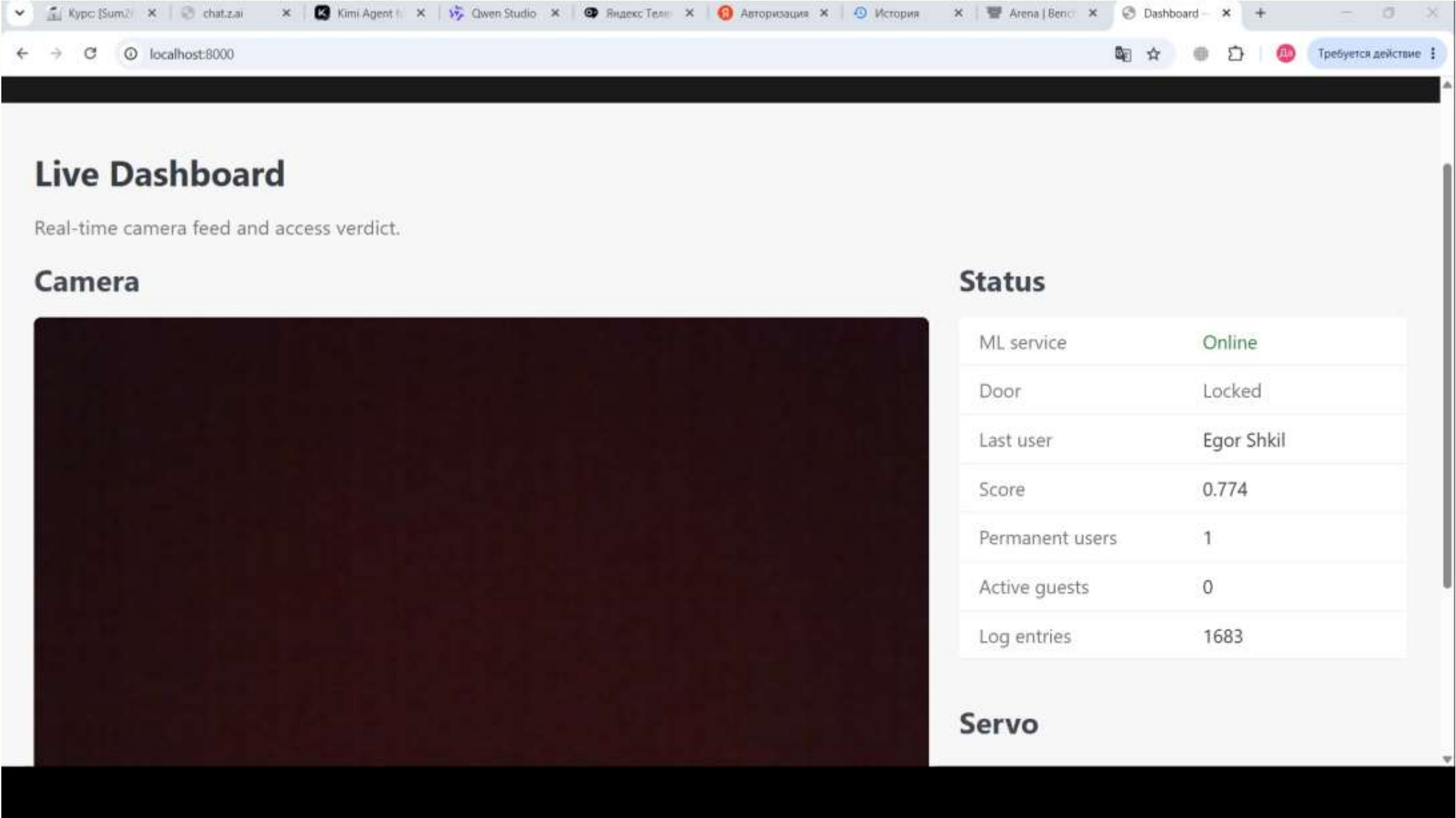
Solution

The system detects a face, compares it against a database of registered users and decides: known or unknown. On successful recognition, a servo motor turns and the door opens.

Stakeholders

- Customer
- Laboratory workers
- Laboratory visitors

Demo



Team

Members	Roles	Contribution
Dmitrii Pozdniakov	Team Lead + Product Owner	Management & Workflow: Supervised the project repository setup, managed GitHub workflows, tracked team issues, and handled final assignment submissions via Moodle. Customer Relations: Conducted formal customer interviews, drafted targeted client meeting questions, and compiled comprehensive project documentation files.

Team

Members	Roles	Contribution
Emil Mardanov	Backend + Developer	Development & CI/CD Support: Formulated the project's official Definition of Done (DoD) and debugged critical pipeline tasks, including fixing the Lychee link checker. Requirements & Analysis: Contributed heavily to initial user story generation, completed interview transcriptions, conducted feature observations, and updated the project's qualitative analysis metrics.

Team

Members	Roles	Contribution
Shamil Gabdrakhmanov	Backend + Developer	Full-Stack MVP Development: Actively developed the entire framework for MVP-v1, including the backend architecture, database schemas, and frontend user interface elements. Documentation & Research: Authored the progress reports for multiple project milestones, conducted market research on similar projects, and analyzed validation results using the "Mom Test" framework.

Team

Members	Roles	Contribution
Nadezhda Voskan	ML-engineer + Scrum Master	Development & Backlog Grooming: Built and integrated the functional machine learning service, drafted user stories, and systematically created and organized technical GitHub issues. Discovery & Notes: Performed extensive competitive analysis across related software projects, managed audio recordings of client interactions, and maintained detailed meeting logs.

Team

Members	Roles	Contribution
Egor Shkil	ML-engineer + Developer	ML Engineering & Prototyping: Supported the core development of both the initial MVP v0 and the current MVP v1, designed the product's early interface prototype, and actively resolved critical bugs on the core MVP. Agile Management: Set up the team's project ticket board, authored the vast majority of the Product Backlog Items (PBIs), and delivered the project's retrospective.md documentation. Client Validation Prep: Outlined initial MVP features, prepared questions for customer validation meetings, and compiled interview transcripts.

Team

Members	Roles	Contribution
Mikhail Brovkin	Hardware engineer + Developer	Hardware Integration: Organized hardware sync meetings and spearheaded the physical deployment, configuration, and testing of MVP v1 on the Raspberry Pi unit. Analysis & Repository Setup: Managed technical repository assignment configurations, drafted evaluation questions, captured demonstration videos, and updated the project matrix regarding artificial intelligence implementations.

Requirements

Functionality	Priorities
Face recognition	Must have
Automatic door unlock when the face is “known”	Must have
Door remains closed when the face is “unknown”	Must Have
All users are stored in the database	Must Have
The type of access either constant or temporary	Must Have
Implementation on Raspberry Pi	Must Have
Liveness detection	Could Have

The main quality is facial recognition and automatic door unlocking and opening

Roadmap

Current state of the project

The project is represented as the MVP v1 that includes face recognition, backend for the admin to perform CRUD operations, the project is ready for use on Raspberry

Plan for the upcoming weeks

Complete the functionality for the “Update” operation to dynamically change type of access of the users

What is in use by the client

The customer has not started using the product yet, since the product must first be fully completed in terms of software and hardware. For further use, the customer will need to install a ready-made system on the door. However, during meetings with the customer, it became clear that the customer is not yet ready to use the product in the near future.

Useful feedback

The customer is generally satisfied with the work done so far, but noted inaccuracies that are indicated in the Roadmap as a plan for further action

Current challenges

New hardware such as Raspberry Pi as well as working with the backend and frontend parts of the project took more time than other parts of the project. Also, the GitHub workflow caused difficulties at first. The team successfully copes with all the challenges.

Links



[Project](#)



[Repository](#)